

EARLY TOOTH DEVELOPMENT

TUTORIAL
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PHASES OF TOOTH DEVELOPMENT (OVERLAPPING PHASES)

Initiation

Morphogenesis

Histogenesis

Tooth development:
(complex interactions
between)

epithelia

mesenchyma
l tissues.



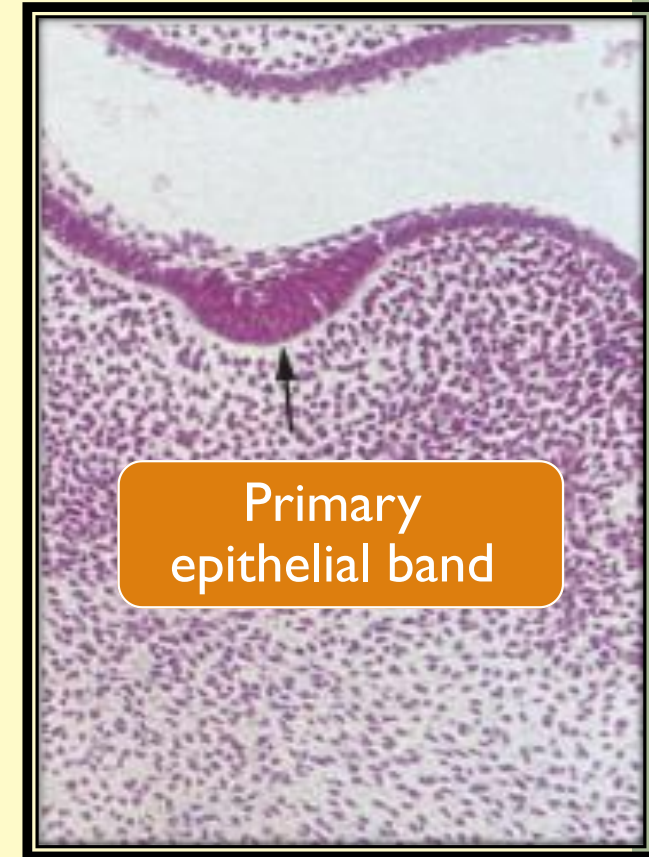
FIRST HISTOLOGICAL SIGN OF TOOTH DEVELOPMENT:

Week : 6th
IU

Appearance of a condensation of mesenchymal tissue and capillary networks beneath the presumptive dental epithelium of the primitive oral cavity.

6th week of
development

The oral epithelium thickens and invaginates into the mesenchyme to form a “**primary epithelial band**”.



Primary
epithelial band

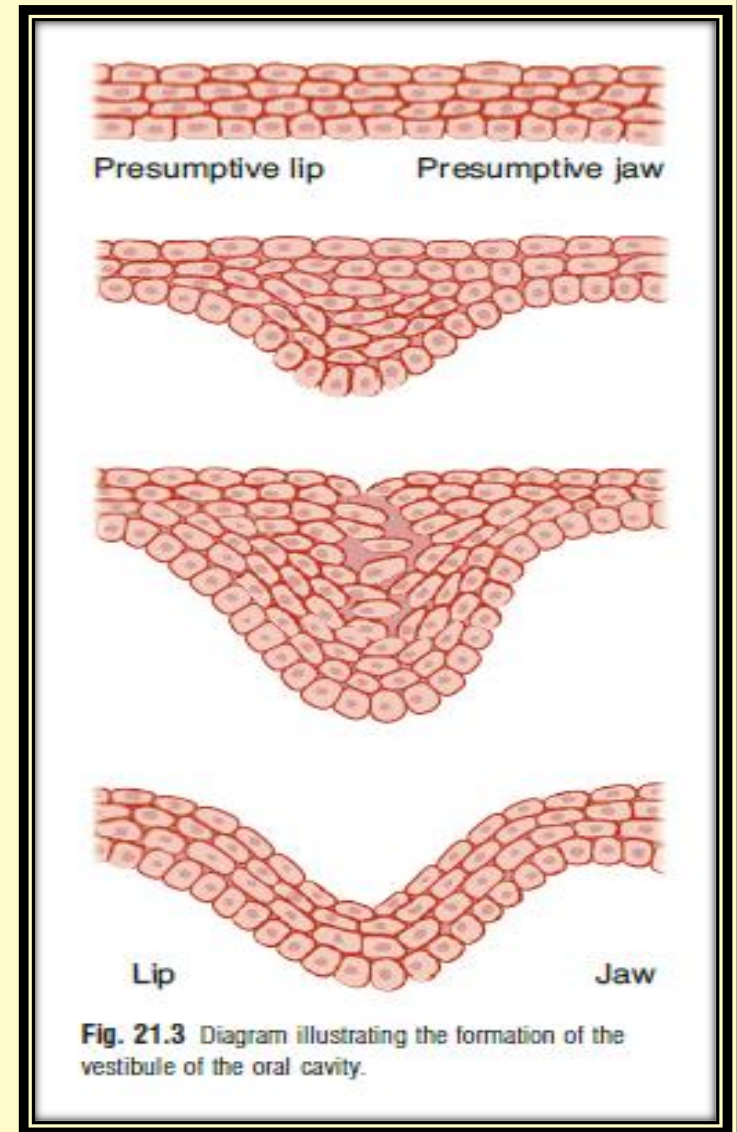
7th week

**Week :
7th IU**



Vestibule formation

- The cells of the vestibular lamina proliferate, with subsequent degeneration of the central epithelial cells to produce the sulcus of the vestibule



8th week

- Series of swellings
- arch-shaped dental lamina.
- epithelial swelling → completely surrounded by mesenchymal condensation.

Week : 8th
IU

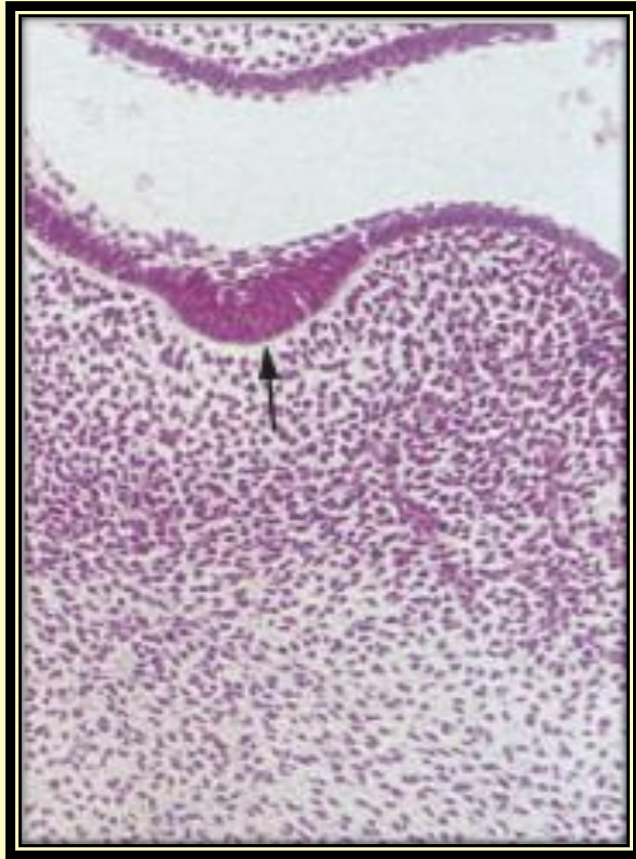


CLASSIFICATION OF TOOTH GERM (DESCRIPTIVE PURPOSES)



According to the degree of morphodifferentiation and histodifferentiation of their epithelial components (enamel organs).

**TASK 1: IDENTIFY, PASTE
AND LABEL THE
FOLLOWING FIGURES**



Primary Epithelial Band

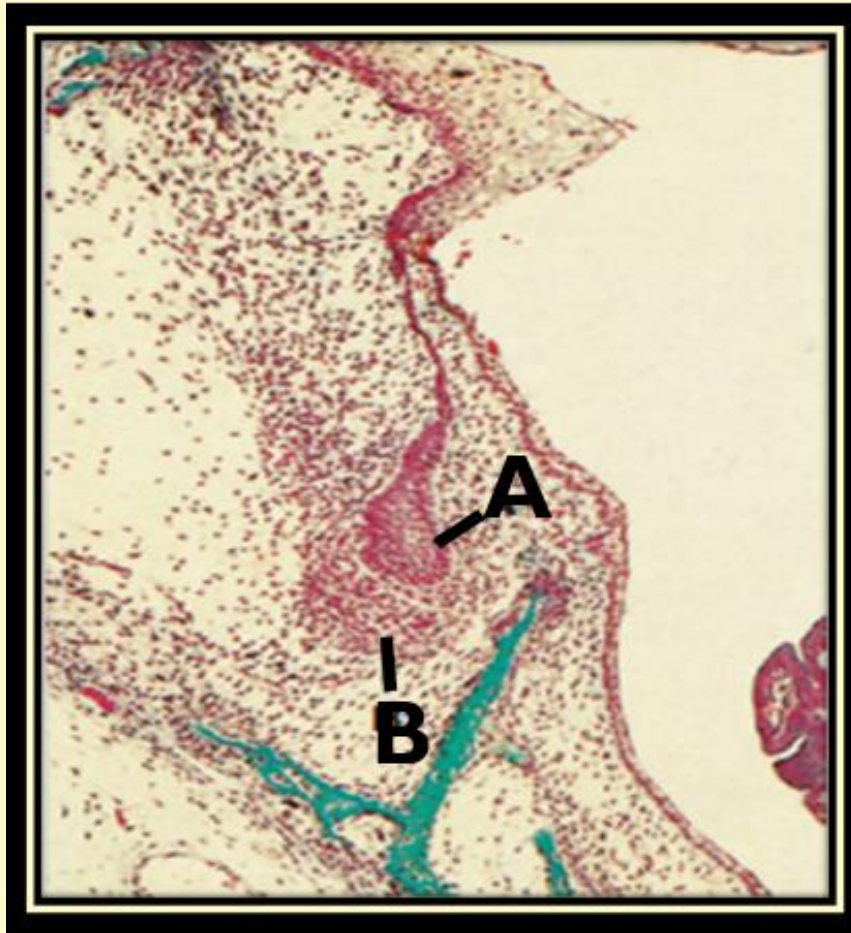
Week : 6th IU

**A :Vestibular Lamina
(Buccally located)**

**B: Dental Lamina
(Lingually located)**



Week : 7th IU



A : Enamel Organ

B: Mesenchymal
Condensation

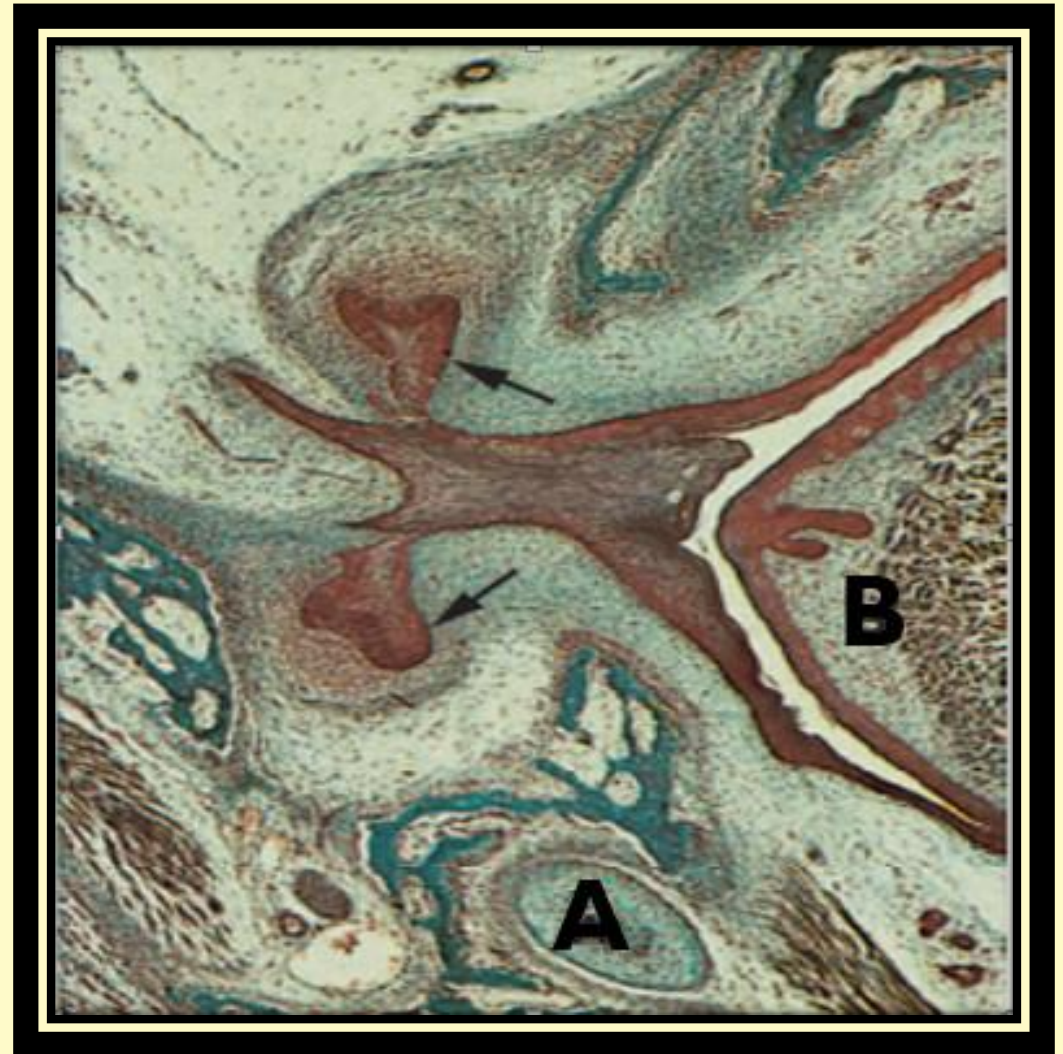
Bud Stage of Tooth Development

Early Cap Stage of Tooth Development

A : Meckel's Cartilage

B: Developing Tongue

Week : 11th IU
(for developing deciduous teeth)





Late Cap Stage of Tooth Development

A : Stellate Reticulum

B : External Enamel Epithelium

C : Internal Enamel Epithelium

D : Dental Papilla

E : Dental Follicle

Week : 12th IU (for developing deciduous teeth)

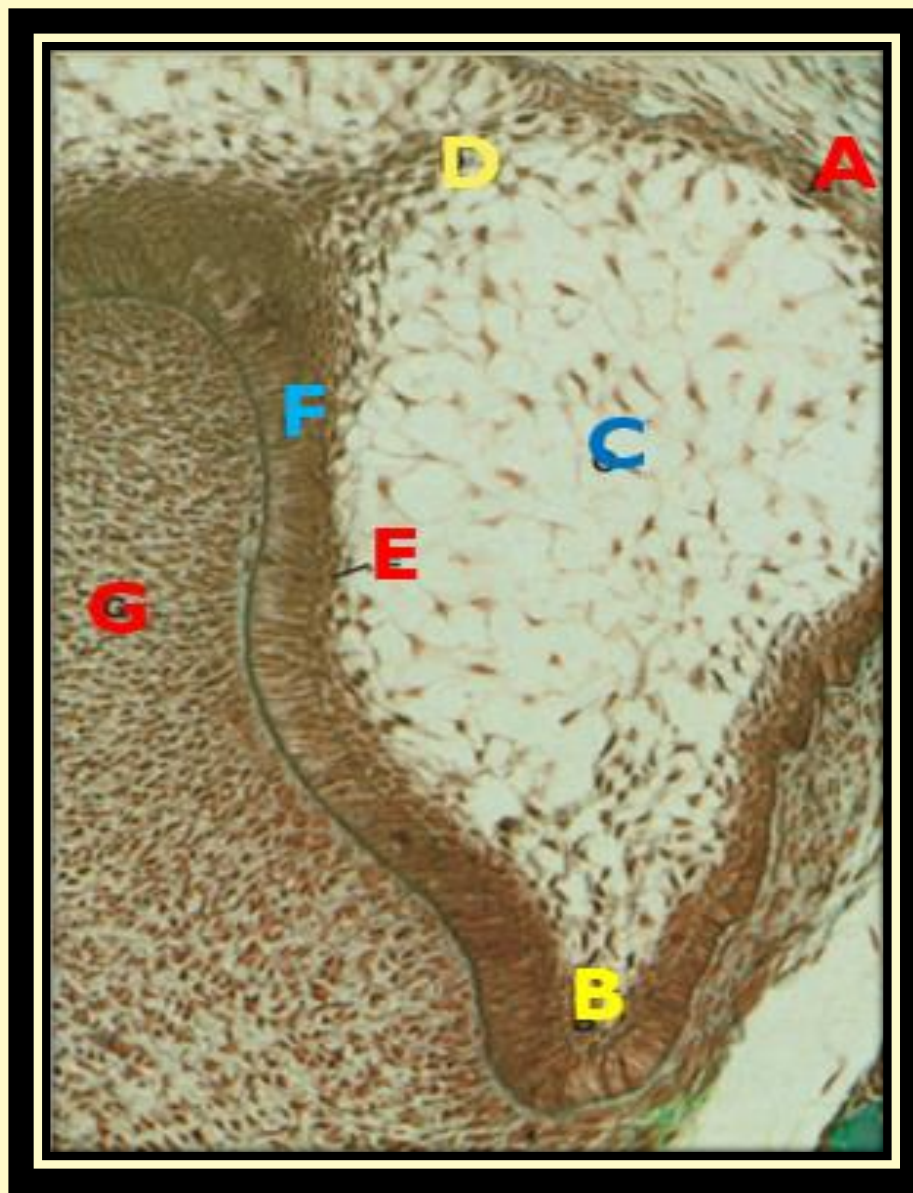
Early Bell Stage of Tooth Development

A : Inner Investing Layer of Dental Follicle

B: Outer Layer of Dental Follicle



Week : 14th IU (for developing deciduous teeth)



Early Bell Stage of Tooth Development (High-power view)

A : External Enamel Epithelium

B : Cervical Loop

C : Stellate Reticulum

D : Enamel Cord

E : Stratum

Intermedium

F : Internal Enamel Epithelium

Late Bell Stage (Appositional Stage) of Tooth Development

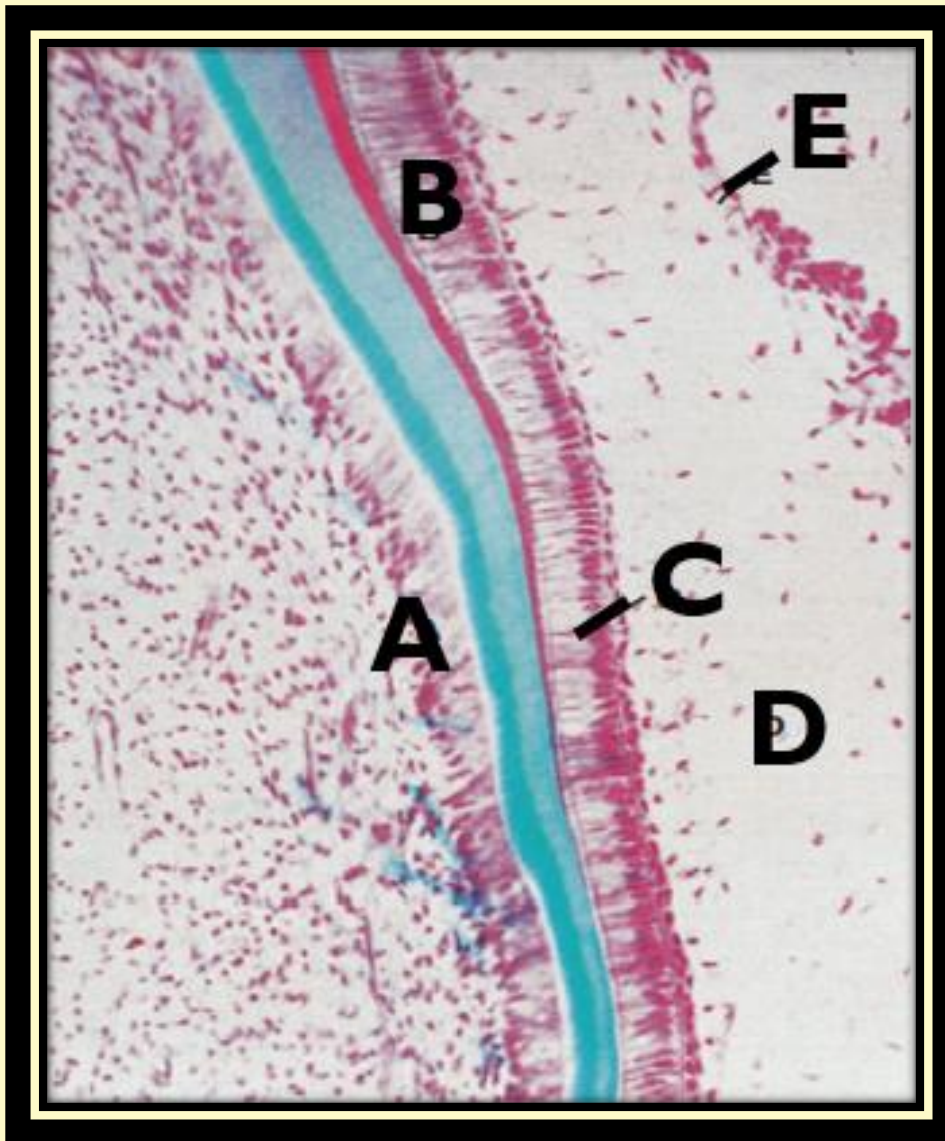
Enamel Matrix

Dentin Matrix

A : permanent tooth



Week : 18th IU (for developing deciduous teeth)



Late Bell Stage (High Power View) of Tooth Development

A : Odontoblasts

B : Ameloblasts

C : Stratum

Intermedium

D : Stellate Reticulum

E : External Enamel

Epithelium



TASK 2: ANSWER THE FOLLOWING QUESTIONS

Q1. WHAT ARE THE DIFFERENCES BETWEEN EARLY AND LATE CAP STAGE.

S. no	Early Cap Stage	Late Cap Stage
1.	Occurs at 11 th week IU for developing deciduous teeth.	Occurs at 12 th week IU for developing deciduous teeth.
2.	Progression of morphodifferentiation, with poor histodifferentiation and deeper surface of enamel organ invaginate to form cap shaped structure	Slightly better histodifferentiation than early cap stage.
3.	Characterized by central rounded cells and peripheral Internal and External Enamel Epithelium.	Cells of stellate reticulum separated by GAG filled in I/C spaces and contact by desmosomes Cells of External Enamel Epithelium, cuboidal Cells of Internal Enamel Epithelium, columnar, rich in RNA, hydrolytic and oxidative enzymes.
4.	Dental papilla and dental follicle can not be appreciated.	Dental papilla and dental follicle can be appreciated.

Q2. WHAT ARE THE DIFFERENCES BETWEEN EARLY AND LATE BELL STAGE.

S. no	Early Bell Stage	Late Bell Stage
1.	Commencement about 14 th week IU for developing deciduous teeth.	Commencement about 18 th week IU for developing deciduous teeth.
2.	Occurs due to further morphodifferentiation and histodifferentiation of the tooth germ	Associated with the formation of the dental hard tissues. Dentine formation always precedes enamel formation
3.	• Involves mapping of the occlusal pattern of the crown of the tooth by Internal enamel epithelium; • loss of contact b/w oral epithelium & enamel organ by degeneration of dental lamina • formation and recognition of dental follicle and its investing, fibrocellular and Vascular layer • histodifferentiation of enamel organ into external enamel epithelium, stellate reticulum, stratum intermedium and internal enamel epithelium; • Cervical loop can be appreciated as a site of mitotic activity	Involves dentinogenesis which forms first dentin and amelogenesis, through reciprocal induction Lingual downgrowths give rise to the tooth germs of the permanent successors

Q3. WHAT IS MEANT BY OCCLUSAL MAPPING OF FUTURE TEETH

- Due to configuration and folding of IEE
- Differential mitosis along IEE.
 - Site of cell maturation & Cessation of mitosis : The future cusps and incisal margins
 - Mitotically active areas: fissures & margins of tooth.
- Increasing cusp height → continued downward growth at tooth margin and fissures rather than upward extension of the cusps.

Q4. WHAT IS RECIPROCAL INDUCTION?

RECIPROCAL INDUCTION

- Inductive influence of developing ameloblasts (preameloblasts) differentiate adjacent mesenchymal cells of the dental papilla into odontoblasts
- Odontoblasts then become involved in the formation of predentine and dentine.
- The presence of dentine then induces the ameloblasts to secrete enamel.

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TASK 3: IDENTIFY AND MATCH THE FOLLOWING FIGURES TO THE CORRECT DESCRIPTION

A



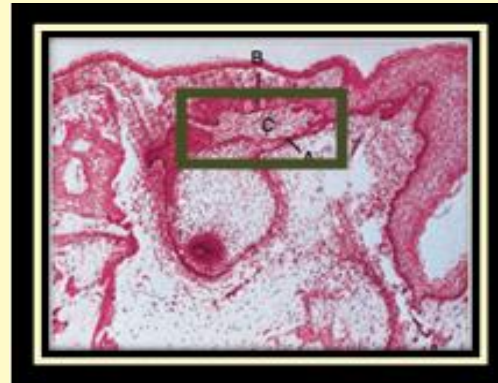
B



C



D



**MATCH THE
LETTERS TO
THE CORRECT
DESCRIPTION
NUMBERS**

1: Enamel

Niche

2: Enamel Cord

3: Enamel knot

4: Enamel Navel

KEY:

A → Enamel Knot : A localised, nonproliferating cluster of cells at the centre of the inner dental epithelium, which is central to establishing the form of the tooth crown and forms a bulge into dental papilla in the centre of enamel organ.

Incisors and Canines (1 primary enamel knot)

Premolars and molars (1 primary Enamel knot, additional secondary enamel knots)

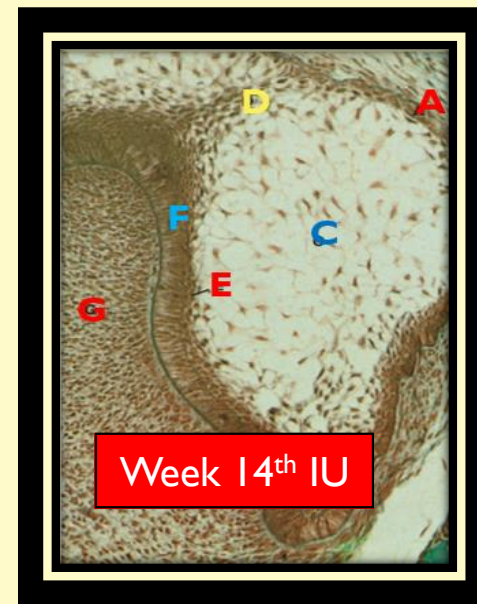
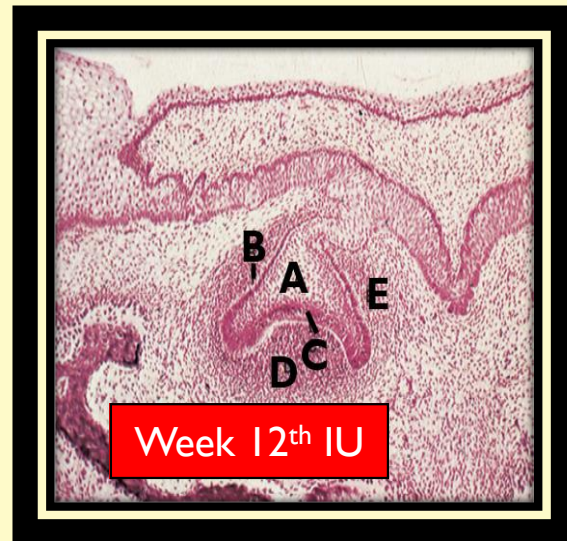
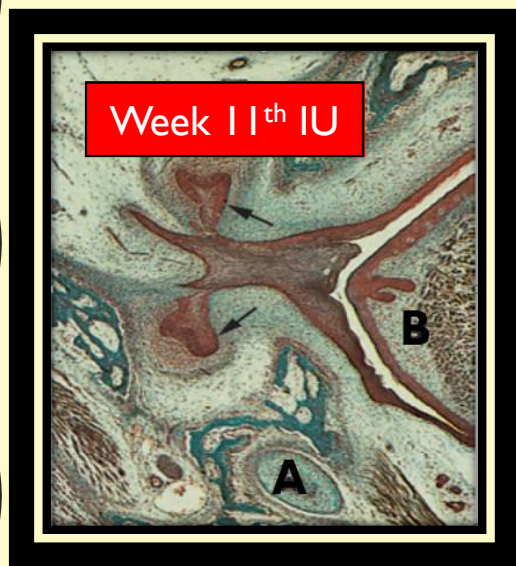
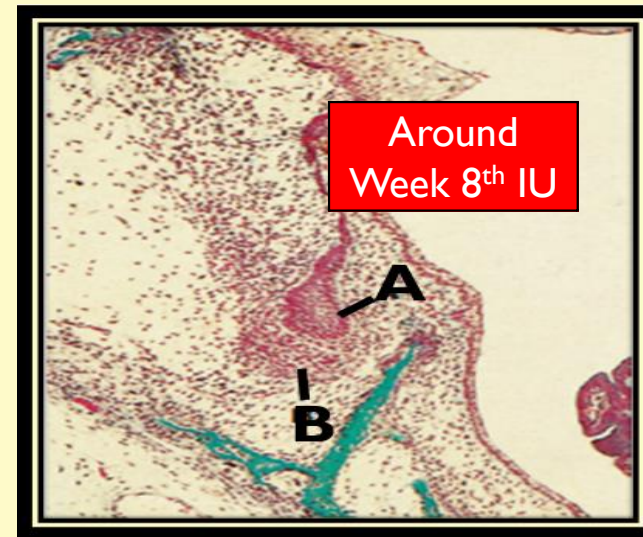
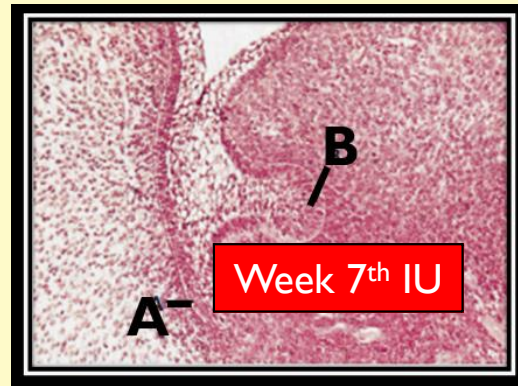
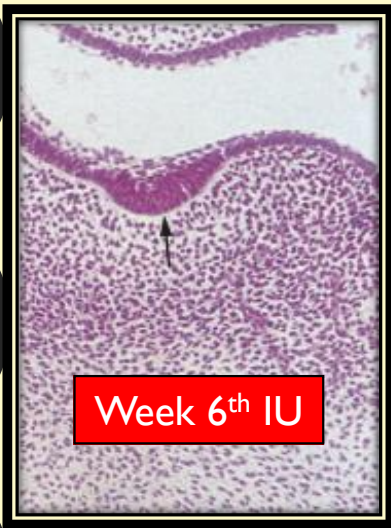
B → Enamel cord : A strand of cells seen at the early bell stage of development that extends from the stratum intermedium into the stellate reticulum.

C → Enamel navel : A small invagination seen where the enamel cord meets the external enamel epithelium.

D → Enamel Niche : It appears as a funnel-shaped depression containing connective tissue, seen where the tooth germ appears to have a double attachment to the dental lamina (the lateral and medial enamel strands).

A decorative graphic on the left side of the slide, consisting of a light yellow wavy line with a green outline, resembling a stylized tooth or a wave.

TASK 3: IDENTIFY EACH STAGE OF TOOTH DEVELOPMENT AND IT'S WEEK OF DEVELOPMENT





EXTRA REFERENCE MATERIAL



Fig. 21.40 Patient with ectodermal dysplasia showing several upper teeth present but with the characteristic peg shape. From *Cawson's essentials of oral pathology and oral medicine*, 8th ed. 2008.



Fig. 21.41 Young patient with ectodermal dysplasia showing a complete absence of teeth. Courtesy Dr S Saraf.



Fig. 2.164 Radiograph showing an inverted, supplemental (4th), permanent molar, high up in the ramus of the mandible. Courtesy Dr B J Doherty.



Fig. 2.166 Upper jaw showing the presence of two mesiodens. Courtesy the Royal College of Surgeons of England.



Fig. 26.55 Orthopantomogram of patient with cleidocranial dysplasia (dysostosis). The majority of teeth are unerupted and may include supernumerary teeth.



Fig. 2.167 An example of microdontia in the form of a 'peg' maxillary lateral incisor (arrow). Courtesy Dr M Cobourne.



Fig. 2.168 An example of macrodontia affecting the maxillary left first incisor.

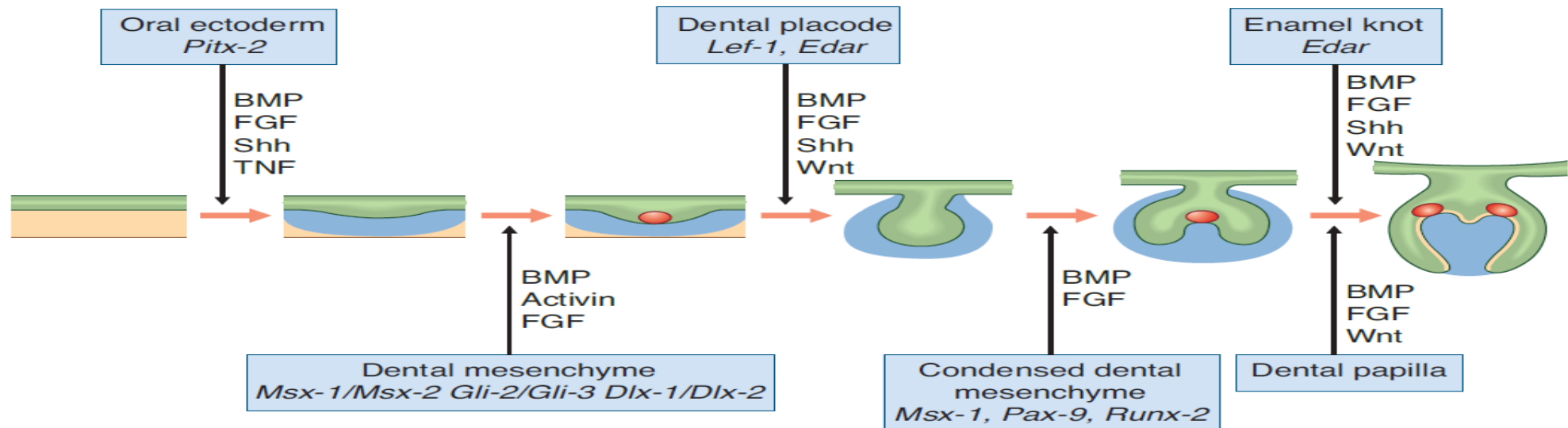


Fig. 21.38 Model of the molecular regulation of tooth development from initiation to crown morphogenesis. Interactions between epithelial (green) and mesenchymal tissues (blue) are mediated by signal molecules (BMP = bone morphogenetic proteins; FGF = fibroblast growth factors; Shh = sonic hedgehog; TNF = tumour necrosis factor). These signals operate throughout development and regulate the expression of genes in the responding tissues (shown in boxes). Signalling centres (red) appear in the epithelium reiteratively and secrete locally many different signals that regulate morphogenesis and tooth shape. Courtesy Prof. I Thesleff.

Genes & Proteins related to different teeth:

Incisor: **MSX I, MSXII, ALX3**
(BMP-4)

Molars: **BARX I** (FGF-8, FGF-9)

Canines and premolars: **MSX I, MSXII**

FIELD MODEL HYPOTHESIS

- Each tooth germ starts the same, but **different concentrations of growth factors (GF)** are responsible for producing different tooth types.
- Patterning of the dentition is related to the **polarity of the epithelium.**

FGF-8, BMP-4

CLONAL MODEL HYPOTHESIS

- Tooth type is prespecified at original site of Neural Crest Cells & is not dependent on environment of jaws.
- Supported by succedaneous tooth formation i.e 2nd and 3rd molars.

Box 21.1 Stem cell populations in teeth

- Dental pulp stem cells (DPSC)
- Periodontal ligament stem cells (PDLSC)
- Stem cells from apical papilla (of erupting and newly erupted teeth) (SCAP)
- Dental follicle stem cells (DFSC)
- Stem cell from exfoliated deciduous teeth (SHED)

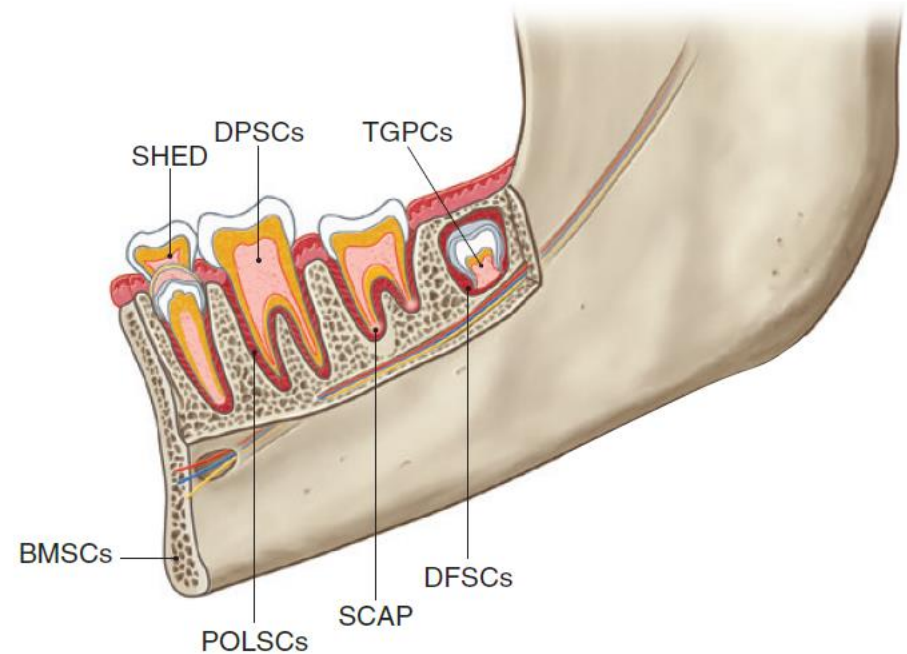


Fig. 21.39 Diagram showing source of stem cells in the regions of the jaws and teeth. Sites of various populations of dental stem cells. BMSC = bone marrow stem cells; DPSC = dental pulp stem cells; SHED = stem cells from the pulp of human exfoliate deciduous teeth; TGPC = stem cells of the tooth germ's papilla cells. Redrawn from Fouad SAA 2015 Dental stem cells: a perspective area in dentistry. *International Journal of Dental Sciences and Research* 3: 15–25.



Fig. 21.43 Two upper molar teeth being bound together by cementum (concrecence). From *Cawson's essentials of oral pathology and oral medicine*, 8th ed. 2008.



Fig. 2.169 (A) Geminated maxillary permanent incisor tooth. (B) Geminated maxillary premolar tooth. (A, B) Courtesy the Royal College of Surgeons of England.

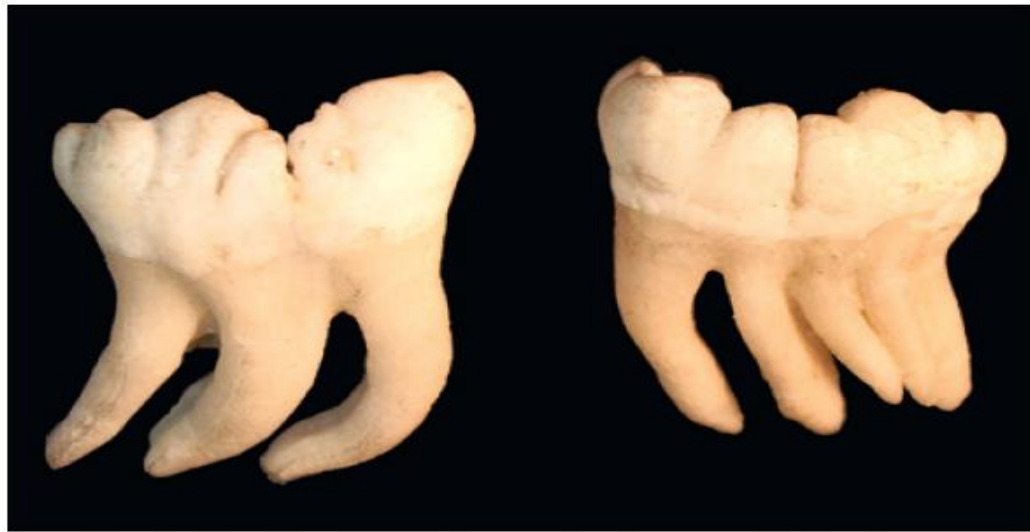


Fig. 2.170 Geminated mandibular permanent molar tooth. Courtesy the Royal College of Surgeons of England.





Fig. 21.44 Maxillary permanent first incisors from a patient with congenital syphilis, showing a barrel-shaped outline and notched edge ('Hutchinson's incisors'). Courtesy the Royal College of Surgeons of England.

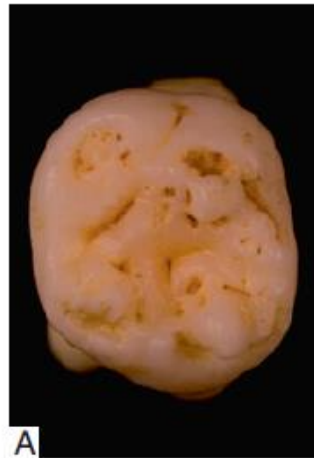


Fig. 21.45 Maxillary permanent molar teeth from a patient with congenital syphilis. (A) Occlusal view and (B) side view, both showing projections on the crown surface ('mulberry molars'). Courtesy the Royal College of Surgeons of England.

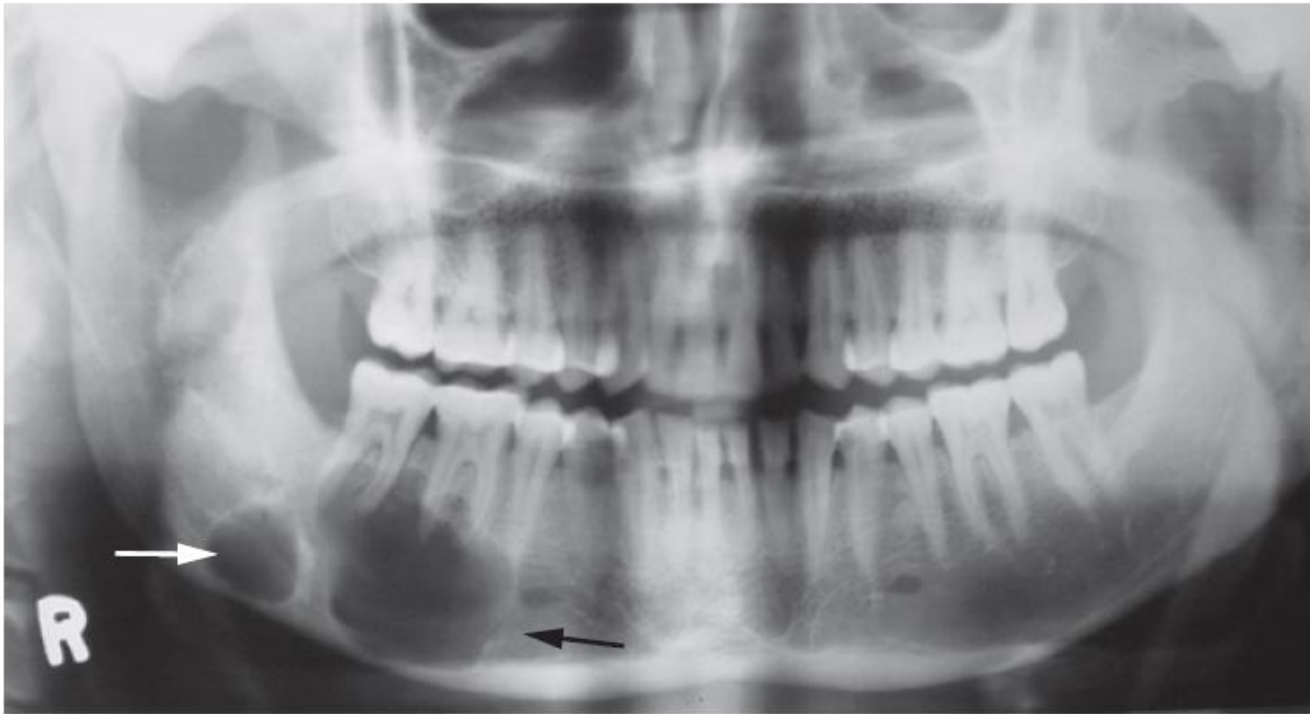


Fig. 21.46 A radiograph showing the presence of an ameloblastoma in the right quadrant of the mandible of a 35-year-old male patient. It may present as a multilocular cyst or as multiple cystic cavities (arrows). Courtesy Dr S Saraf.

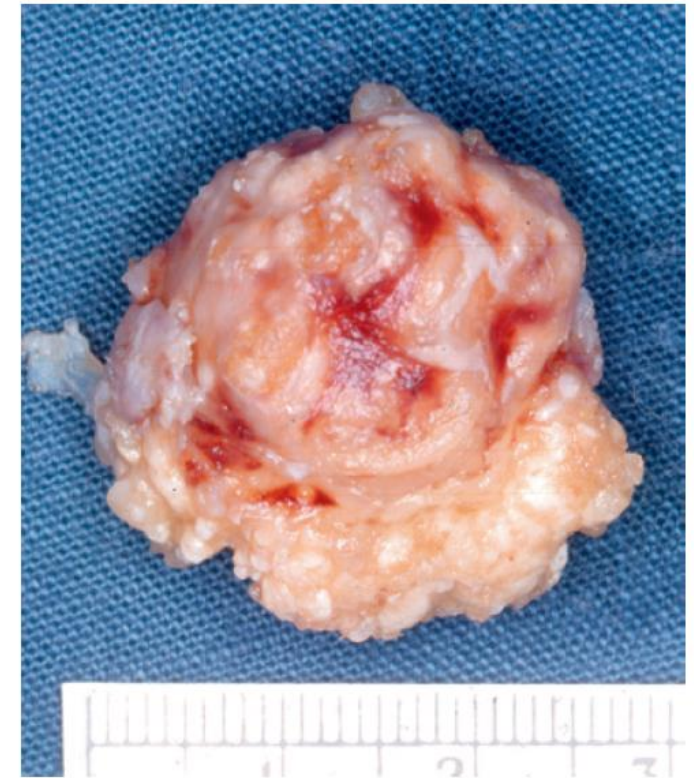


Fig. 21.52 Complex odontome removed from a patient revealed as a calcific mass. There is a low degree of morphodifferentiation compared with the compound odontome seen in Fig. 21.50. Courtesy Dr J D Harrison.

*Thank
you!*